SARA BAKHTIARI

PhD Candidate in Mechanical Engineering | Computational Mechanics | Scientific Computing | Machine Learning & AI

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PROFILE

Mechanical Engineering PhD Candidate (GPA 3.91/4.00) developing nonlinear finite-element and multiphysics models for cardiovascular mechanics, with experience in scientific computing, HPC, model validation, data analysis, and AI/ML for engineering applications.

TECHNICAL SKILLS

- **Programming & Scientific Computing:** Python, MATLAB, NumPy, SciPy, Pandas, FEniCS/DOLFIN, UFL
- **Machine Learning & AI:** scikit-learn, vision-language models, embeddings, regression, cross-validation, feature selection, dimensionality reduction, learned bottlenecks
- **Computational Mechanics & HPC:** nonlinear FEA, continuum mechanics, constitutive modeling, anisotropic soft-tissue mechanics, MPI, mpi4py, PETSc, SLURM, Linux, Singularity
- **Visualization & Engineering:** ParaView, Matplotlib, COMSOL Multiphysics, SolidWorks, HDF5/XDMF/CSV/NumPy workflows

EDUCATION

PhD, Mechanical Engineering (Direct PhD) - University of Kentucky **Spring 2024-Present**

Candidate since Apr 2026 | GPA 3.91/4.00 | Advisor: Jonathan F. Wenk, PhD

B.Sc., Mechanical Engineering - Iran University of Science and Technology (IUST) **2016-2021**

RESEARCH EXPERIENCE

Graduate Research Assistant - Computational Cardiac Mechanics **Spring 2024-Present**

University of Kentucky - Lexington, KY

- **MyoFE_H - Multiphysics Cardiac Mechanics & Remodeling:** Develop and extend a coupled FEA framework for ventricular mechanics, passive constitutive laws, fiber disarray, growth/remodeling, and agent-based fibrosis, integrating half-sarcomere kinetics, active contraction, and cardiovascular circulation.
- **Simulation, Validation & HPC:** Run large-scale Python/FEniCS simulations with MPI, PETSc, SLURM, and Linux; build validation and post-processing workflows for material tests, ventricular inflation, stress/strain fields, pressure-volume loops, energetics, and ParaView.

Research Project - Machine Learning & Vision-Language Models for Electrospinning **2026**

- **AI/ML Pipeline:** Applied pretrained and fine-tuned vision-language models to electrospinning videos; evaluated embeddings, learned bottlenecks, feature variability/selection, regression performance, repeatability, and repeated cross-validation for fiber-diameter prediction.

SELECTED PUBLICATIONS & RESEARCH OUTPUTS

Peer-reviewed: Islam, R., Bakhtiari, S., Anugrah, G. G., Mehri, M., & Wenk, J. F. "Advancements in finite element modeling of myocardial infarction: a review." Biomechanics and Modeling in Mechanobiology, 25(4), 2026. DOI: 10.1007/s10237-026-02107-0
Under review: "Choice of Passive Myocardial Constitutive Law Drives Divergent Predictions of Left Ventricular Remodeling in Volume-Overload Conditions"; "Effect of Myocardial Fiber Disarray on Left Ventricular Mechanical Efficiency and Baroreflex Compensation."
Conference: BMES Annual Meeting, San Diego, CA, Oct. 2025 - passive constitutive law effects on LV mechanics and growth under disease conditions.

TEACHING

Teaching Assistant - Statics, University of Kentucky (Fall 2026) | Teaching Assistant - Statics, IUST (2021)